

Category Theory, Sheaf and Presheaf

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1 Category, Functor and Natural Transformation

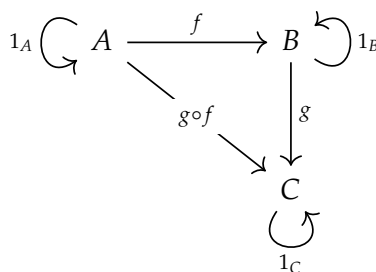


Figure 1: Category

Category

Definition 1. A **category** C consists of the following data:

- (i) A class $\text{Ob}(C)$, called *objects* of category C .
- (ii) For a pair A and B of objects, a class $\text{Mor}_C(A, B)^a$ whose elements are called *morphisms* (or *arrows*) from A to B .
- (iii) For a pair of morphisms f in $\text{Mor}_C(A, B)$ and g in $\text{Mor}_C(B, C)$, there is a *composition* $g \circ f$ in $\text{Mor}_C(A, C)$.

These data are to satisfy the following rules:

- (i) For every object A , there is an identity morphisms 1_A in $\text{Mor}_C(A, A)$.
- (ii) Composition is associative.

^aSometimes, we also denote it by $\text{Hom}(A, B)$ or $C(A, B)$.

Remark 1.

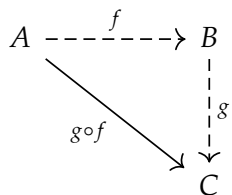


Figure 2: Composition of Morphisms

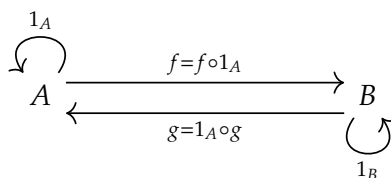


Figure 3: Identity Morphisms

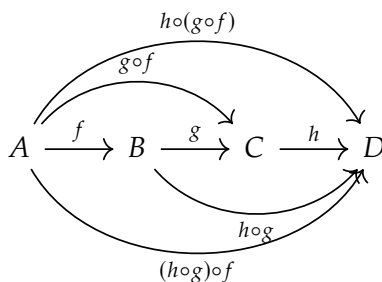


Figure 4: Associativity of Composition

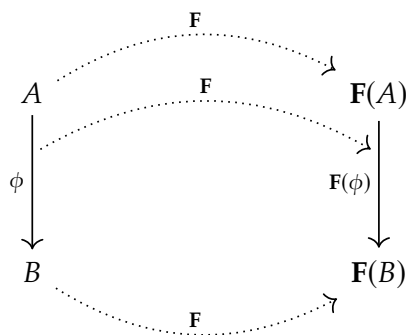
Functor

Definition 2. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between two categories $\mathcal{C}$ and $\mathcal{D}$ is given by the following data:

- (i) A map $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$
- (ii) For every A and B in $\text{Ob}(\mathcal{C})$, a map F given by

$$\begin{aligned} F : \text{Mor}_{\mathcal{C}}(A, B) &\longrightarrow \text{Mor}_{\mathcal{D}}(F(A), F(B)) \\ \phi &\longmapsto F(\phi) \end{aligned}$$

Remark 2.



Natural Transformation

Definition 3. Let $F, G : C \rightarrow D$ be functors. A **natural transformation** (or a **morphism of functors**) $\eta : F \rightarrow G$, is a collection $\{\eta_A\}_{A \in \text{Ob}(C)}$ such that

- (i) $\eta_A : F(A) \rightarrow G(A)$ is a morphism in the category D , and
- (ii) for each morphism $\phi : A \rightarrow B$ of C the following diagram is commutative

$$\begin{array}{ccc} F(A) & \xrightarrow{\eta_A} & G(A) \\ F(\phi) \downarrow & & \downarrow G(\phi) \\ F(B) & \xrightarrow{\eta_B} & G(B) \end{array}$$

Sometimes we use the diagram

$$\begin{array}{ccc} & \mathbf{F} & \\ \curvearrowright & \Downarrow \eta & \curvearrowleft \\ C & & D \\ \curvearrowleft & \Uparrow \mathbf{G} & \end{array}$$

to indicate that η is a morphism from F to G .

Remark 3.

$$\begin{array}{ccccc} & & & \mathbf{G} & \\ & & & \cdots & \\ & & & \curvearrowright & \\ A & \cdots \cdots \cdots \mathbf{F} \cdots \cdots \cdots & F(A) & \xrightarrow{\eta_A} & G(A) \\ & \vdots & \downarrow \psi = F(\phi) & & \downarrow \chi = G(\phi) \\ & \cdots \cdots \cdots \mathbf{F} \cdots \cdots \cdots & & & \\ \phi \downarrow & \cdots \cdots \cdots \mathbf{G} \cdots \cdots \cdots & G & \cdots \cdots \cdots & \\ B & \cdots \cdots \cdots \mathbf{F} \cdots \cdots \cdots & F(B) & \xrightarrow{\eta_B} & G(B) \\ & \vdots & \downarrow & & \downarrow \\ & \cdots \cdots \cdots \mathbf{G} \cdots \cdots \cdots & & & \\ & & & \curvearrowleft & \\ & & & \cdots & \end{array}$$

Example 1 (Determinant). TBA

2 Sheaf and Presheaf

Sheaf

Definition 4. TBA

Presheaf

Definition 5. TBA